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> # Атака на передачу связанных сообщений с малым открытым ключом
> with(numtheory) :
> with(RandomTools) :
> e := 3;
                                     e := 3
(1)
=
> p := nextprime(Generate(integer(range=2..10^10)));
                                     p := 5244301331
(2)
=
> q := nextprime(Generate(integer(range=2..10^10)));
                                     q := 9913502003
(3)
=
> n := p·q;
                                     n := 51989391749204065993
(4)
=
> phi_n := (p - 1)·(q - 1);
                                     phi_n := 51989391734046262660
(5)
=
> igcd(e, phi_n);
                                     1
(6)
=
> # Алиса
m1 := Generate(integer(range=2..10^10)) mod n;
a := Generate(integer(range=2..10^5)) mod n;
b := Generate(integer(range=2..10^5)) mod n;
m2 := a·m1 + b mod n;
                                     m1 := 809094428
                                     a := 28001
                                     b := 96274
                                     m2 := 22655453174702
(7)
=
> c1 := m1&^e mod n;
c2 := m2&^e mod n;
                                     c1 := 13989886059368977758
                                     c2 := 16698974375341268905
(8)
=
> # По формуле
M := 
$$\frac{(b \cdot (c2 + 2 \cdot a^3 \cdot c1 - b^3)) \bmod n}{(a \cdot (c2 - a^3 \cdot c1 + 2 \cdot b^3)) \bmod n} \bmod n;$$

                                     M := 809094428
(9)
=
> M - m1;
                                     0
(10)
>

```